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ASSIGNMENT NO: PC – CS 692 / 05

a) Get the title and publisher names of all books. (EQUI JOIN)

```
SQL> select c.title,p.name
  2  from catalog c, publisher p
  3  where c.p_id=p.p_id;

TITLE                                                      NAME
---------------------------------------------------------- ----------------------------
201 Principles of Software Development                     McGraw-Hill
A Guide to SCM                                             Artech House
Airport                                                    Corgi Books
Beachcombing at Miramar                                    Warner Books
Beyond Soap, Water and Comb                                Abbeville Press Publishers
Biplane                                                    Dell Publishing
Business Notes                                             Clarkson N. Porter
Catch 22                                                   Random House
Death March                                                Prentice Hall PTR
Digital Economy                                            McGraw-Hill
Doctors                                                    Bantam Books

TITLE                                                      NAME
---------------------------------------------------------- ----------------------------
Dragon                                                     Harper Collins
First You have to Row a Little Boat                        Warner Books
Hotel                                                      Corgi Books
Illusions                                                  Dell Publishing
Inca Gold                                                  Harper Collins
Jesus CEO                                                  Hyperion
Jesus in Blue Jeans                                        Hyperion
Kane and Abel                                              Harper Collins
Management Mole                                            Corgi Books
Managing Technical People                                  Addison-Wesley
Managing the Software Process                              Addison-Wesley

TITLE                                                      NAME
---------------------------------------------------------- ----------------------------
Silicon Samurai                                            Blackwell Publishers
The Class                                                  Bantam Books
The Complete Adventures of Feluda                          Penguin Books
The End of Imagination                                     DC Books
The Hunt for Red October                                   Harper Collins
```

b) Get the title and publisher names of all books that are priced above 1000. (NON-EQUI JOIN / GREATER-THAN JOIN)

```
SQL> Select c.title,p.name
  2  from catalog c, publisher p
  3  where c.p_id=p.p_id
  4  and c.price>1000;

TITLE                                                      NAME
---------------------------------------------------------- --------------------
A Guide to SCM                                             Artech House
```

c) Get the title, author name, country and price of all the books with India-based authors and price less than 500. (NON_EQUI JOIN / LESS_THAN JOIN)

```
SQL> SELECT c.title, a.name, a.country, c.price
  2  FROM catalog c, author a
  3  WHERE c.a_id = a.a_id
  4  AND a.country = 'India'
  5  AND c.price < 500;

TITLE                                                      NAME          COUNTRY                      PRICE
---------------------------------------------------------- ------------- ---------------------------------------------- -------
The Complete Adventures of Feluda                          Satyajit Ray  India                                            220
The End of Imagination                                     Arundhati Roy India                                            350
The Legends of Khasak                                      O.V. Vijayan  India                                            195
Computers for Everyone                                     Alexis Leon   India                                            200
Countdown 2000                                             Alexis Leon   India                                            275
E-Biz Primer                                               Mathews Leon  India                                            325
ERP Demystified                                            Alexis Leon   India                                            397
Internet for Everyone                                      Mathews Leon  India                                            320
Introduction to Computers                                  Mathews Leon  India                                            180

9 rows selected.
```

d) Get the details of all the books with their categorical descriptions that are priced above 1000. (NATURAL JOIN)

```
SQL> SELECT DISTINCT cat.description
  2  FROM category cat, catalog c
  3  WHERE cat.c_id = c.c_id
  4  AND c.price > 1000;

DESCRIPTION
------------------------------------
Computer Science
```

e) Find out the titles that have the same price. (SELF JOIN)

```
SQL> SELECT c1.title, c2.title, c1.price
  2  FROM catalog c1, catalog c2
  3  WHERE c1.price = c2.price
  4  AND c1.b_id <> c2.b_id;

TITLE                                                    TITLE                PRICE
-------------------------------------------------------- -------------------------------------------------------------- -------
Hotel                                                    Airport        175
Airport                                                  Hotel          175
Silicon Samurai                                          Inca Gold      290
Jesus in Blue Jeans                                      Inca Gold      290
Silicon Samurai                                          Jesus in Blue Jeans                                                290
Inca Gold                                                Jesus in Blue Jeans                                                290
The Class                                                Kane and Abel  380
Jesus in Blue Jeans                                      Silicon Samurai 290
Inca Gold                                                Silicon Samurai 290
Kane and Abel                                            The Class      380

10 rows selected.
```

f) Get the title, author name, publisher name and category name of all books that are published after 2008.

```
SQL> SELECT c.title, a.name, p.name, cat.description
  2  FROM catalog c, author a, publisher p, category cat
  3  WHERE c.a_id = a.a_id
  4  AND c.p_id = p.p_id
  5  AND c.c_id = cat.c_id
  6  AND c.year > 2008;

no rows selected
```

g) Get the details of order which was ordered by a member. (INNER JOIN)

```
SQL> SELECT o.*
  2  FROM order_summary o, member m
  3  WHERE o.m_id = m.m_id;

      O_ID M_ID  ORDER_DAT     AMOUNT ORDER_STATUS
---------- ----- --------- ---------- ---------------
  11096733 C01   25-FEB-00     2038.3 Shipped
  11364223 C03   07-APR-00     1237.5 Shipped
  12806544 C05   30-APR-00      159.5 Shipped
  13976453 C03   25-JUL-00       3575 Cancelled
  26970379 C01   25-JUL-00     3897.3 Shipped
  36532264 C05   22-AUG-00     7012.5 Shipped
  36552341 C01   22-AUG-00     4371.4 Shipped
  55946511 C03   02-FEB-01       1793 Processing
  86464430 C01   07-FEB-01     2131.8 Processing
  86464431 C01   21-FEB-01      982.3 Processing
```

h) By a SQL statement, show how LEFT OUTER JOIN works.

```
SQL> SELECT *
  2  FROM catalog c
  3  LEFT JOIN author a
  4  ON c.a_id = a.a_id;

B_ TITLE                                                   A_ P_ C_       YEAR      PRICE A_ NAME                                                CITY
-- ---------------------------------------------------------- -- -- -- ---------- ---------- -- ------------------------------------------------- -------
-----------------------
COUNTRY
------------------------------
A1 201 Principles of Software Development                  AD MG CS       1995        898 AD Alan M. Davis                                   Colorad
o
USA

E2 ERP Demystified                                         AL TM BS       2000        397 AL Alexis Leon                                     Cochin
India
```

i) By a SQL statement, show how RIGHT OUTER JOIN works.

```
SQL> SELECT *
  2  FROM catalog c
  3  RIGHT JOIN author a
  4  ON c.a_id = a.a_id;

B_ TITLE                                                   A_ P_ C_       YEAR      PRICE A_ NAME                                                CITY
-- ---------------------------------------------------------- -- -- -- ---------- ---------- -- ------------------------------------------------- -------
-----------------------
COUNTRY
------------------------------
A1 201 Principles of Software Development                  AD MG CS       1995        898 AD Alan M. Davis                                   Colorad
o
USA

A2 A Guide to SCM                                          AL AH CS       2000       3715 AL Alexis Leon                                     Cochin
India
```

j) By a SQL statement, show how FULL OUTER JOIN works.

```
SQL> SELECT *
  2  FROM catalog c
  3  FULL OUTER JOIN author a
  4  ON c.a_id = a.a_id;

B_ TITLE                                                   A_ P_ C_       YEAR      PRICE A_ NAME                                                CITY
-- ---------------------------------------------------------- -- -- -- ---------- ---------- -- ------------------------------------------------- -------
-----------------------
COUNTRY
------------------------------
A1 201 Principles of Software Development                  AD MG CS       1995        898 AD Alan M. Davis                                   Colorad
o
USA
```

k) By a SQL statement, show how CARTESIAN PRODUCT works.

```
SQL> SELECT c.*, a.*
  2  FROM catalog c, author a;

B_ TITLE                                                   A_ P_ C_       YEAR      PRICE A_ NAME                                                CITY
-- ---------------------------------------------------------- -- -- -- ---------- ---------- -- ------------------------------------------------- -------
-----------------------
COUNTRY
------------------------------
I4 Introduction to Computers                               ML LV CS       1999        180 WH Watts S. Humphrey                               Florida
USA


988 rows selected.
```

l) Get the details of all authors and publishers in India ordered by name.

```
SQL> SELECT name
  2  FROM author
  3  WHERE country = 'India'
  4  UNION
  5  SELECT name
  6  FROM publisher
  7  WHERE country = 'India'
  8  ORDER BY name;

NAME
--------------------------------
Alexis Leon
Arundhati Roy
Dell Publishing
Mathews Leon
McGraw-Hill
O.V. Vijayan
Prentice Hall PTR
Satyajit Ray
Tata McGraw-Hill

9 rows selected.
```